



The Infected Hour

A Cooperative Action-Public Health Game

CSE 4402: Visual Programming Lab

Team Members



Sadnan Kibria

230041119

Abrar Faiyaz

230041143

Ashique Khan

230041153

The Problem: Why This Game?

THE GAP IN EXISTING GAMES

Most epidemic games treat outbreaks as zombie scenarios focused purely on combat, missing the educational value of real outbreak response.

WHAT PLAYERS NEED

Games that teach real-world public health concepts through engaging cooperative gameplay.

OUR SOLUTION

A 2D cooperative game where players experience outbreak response firsthand while learning about isolation, sanitation, sample collection, and containment.

KEY ISSUES

- Combat-only epidemic games miss the public health health dimension
- No games combine co-op gameplay with health education
- Outbreak concepts remain abstract to general audiences
- Existing games lack emotional narrative depth
- Opportunity to align gaming with SDG 3: Good Health

Goals & Objectives

PRIMARY GOAL: Build a complete 2D cooperative action game with public health mechanics using JavaFX

Dual-Character Gameplay

Implement unique abilities for Field Medic and Local Scout with distinct roles

3+ Progressive Levels

Design levels with escalating difficulty and diverse public health missions

Emotional Narrative

Create visual story panels that reveal character backstory between levels

Cooperative Boss Fight

Develop the Virus Heart encounter requiring requiring genuine two-player cooperation

Polished GUI & HUD

Health bars, contamination meters, inventory system, and mini-map

Fully Playable Deliverable

Complete game with all core mechanics, story, and co-op boss fight

Story & Characters

THE SETTING

A remote village has been infected by the **Virus Heart** — a mysterious central infection source spreading contamination.

THE NARRATIVE HOOK

After every level, a **visual story panel** reveals the protagonist's past.

The village was their childhood home. Their family was affected by an earlier outbreak.

The game transforms from a mission into a personal journey of redemption.



Elric— Field Medic

Healing abilities, cure deployment, and contamination resistance. Carries the weight of a past outbreak that affected his family. The village was his childhood home.



Jane— Local Scout

Knows the village terrain. Excels at rescuing villagers, activating sanitation points, and navigating toxic zones. Grew up in these streets.

Level Objectives & Missions

Each level combines combat with public-health objectives



Isolate Infected Zones

Barricade areas to stop the spread of contamination
contamination



Collect Samples

Gather viral samples for cure research and
deployment



Deliver Medicine

Transport supplies to quarantined villagers in need



Rescue Villagers

Evacuate survivors before contamination spreads
further



Activate Sanitation

Cleanse areas to create safe zones for villagers



Avoid Contamination Clouds

Dynamic toxic zones that expand over time — stay
clear



LEVEL PROGRESSION

Early levels teach mechanics individually. Later levels combine multiple objectives simultaneously, requiring strategic coordination between both players.

The Virus Heart — Final Boss

The central infection source — not just a monster, but a living epidemic.

PROTECTION SYSTEM

- ⚙️ Mutated cell enemies swarm the arena
- 💀 Spreading toxic zones limit movement
- 📶 Infection waves expand across the arena

COOPERATIVE MECHANICS

Elric (PLAYER 1)

Collect and place cure samples to weaken the shield



Jane (PLAYER 2)

Protect villagers and control infected enemies

FAILURE: Contamination reaches 100% = village lost

BOSS PHASES

- 1 Shield Phase
Elric places cure samples while Jane defends against mutations

- 2 Exposure Phase
Shield drops — both players attack the exposed core together

- 3 Core Destruction
Final coordinated assault to destroy the Virus Heart and save the village

VICTORY: COORDINATE TO DESTROY THE VIRUS HEART

Key Features



Local Co-op — Dual-character gameplay with unique abilities



3+ Levels — Progressively challenging mission combinations



Story Panels — Emotional narrative between levels



Contamination — Dynamic toxic cloud expansion system



Interactive World — Barricades, stations, supply crates



Health HUD — Real-time health and contamination bars



Inventory — Manage samples, medicine, and tools



Mini-map — Objectives, contamination, villagers



Boss Fight — Multi-phase cooperative encounter



Audio — Sound effects and atmospheric music

CORE INNOVATION: The only 2D co-op game combining action gameplay with genuine public health education

Technical Tools & Framework

Java 21 

Java 21+

the core programming language



Springboot

Backend logic and database management



Java FX

UI Design



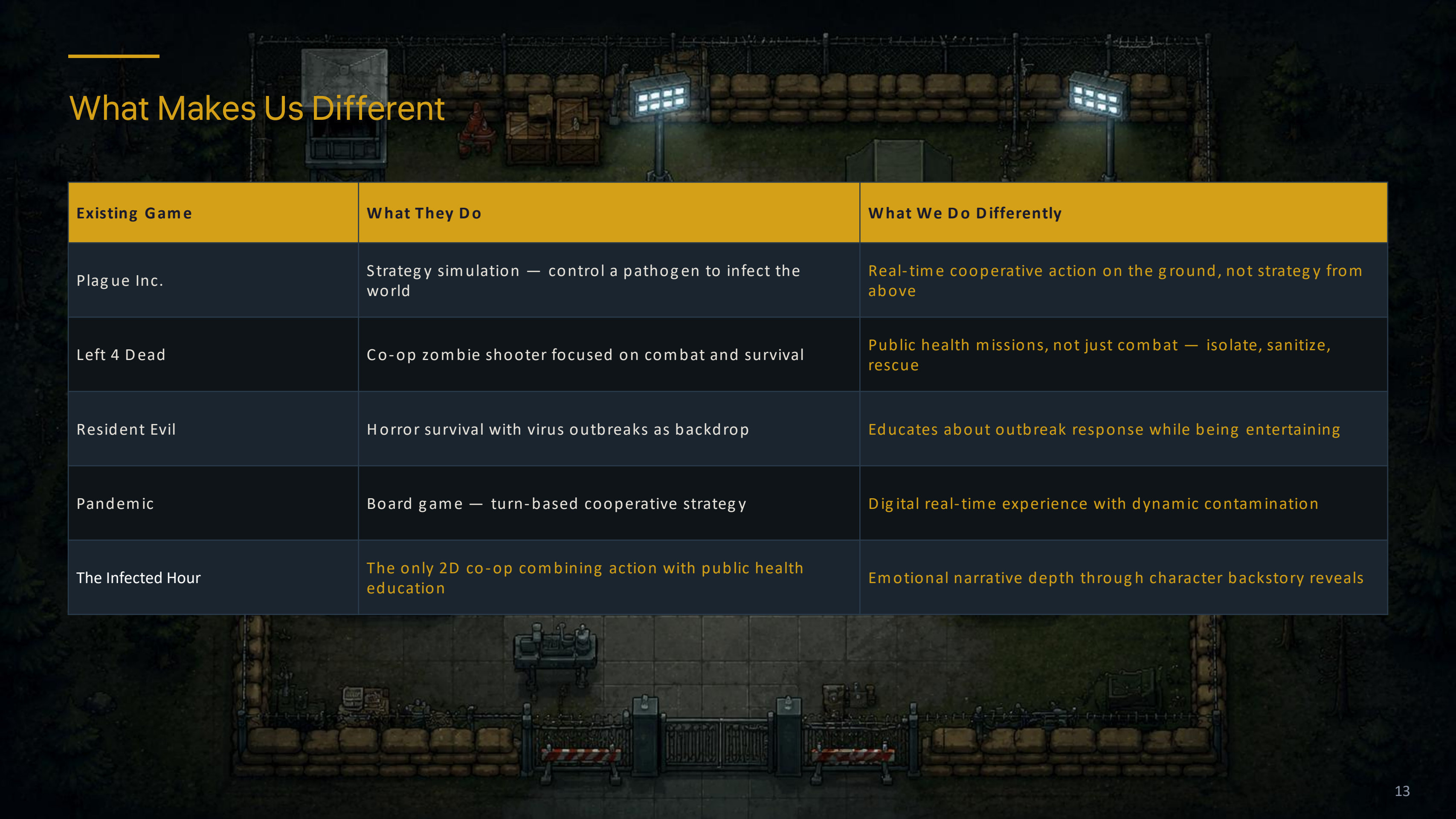
libGDX

2D game development



Git + Github

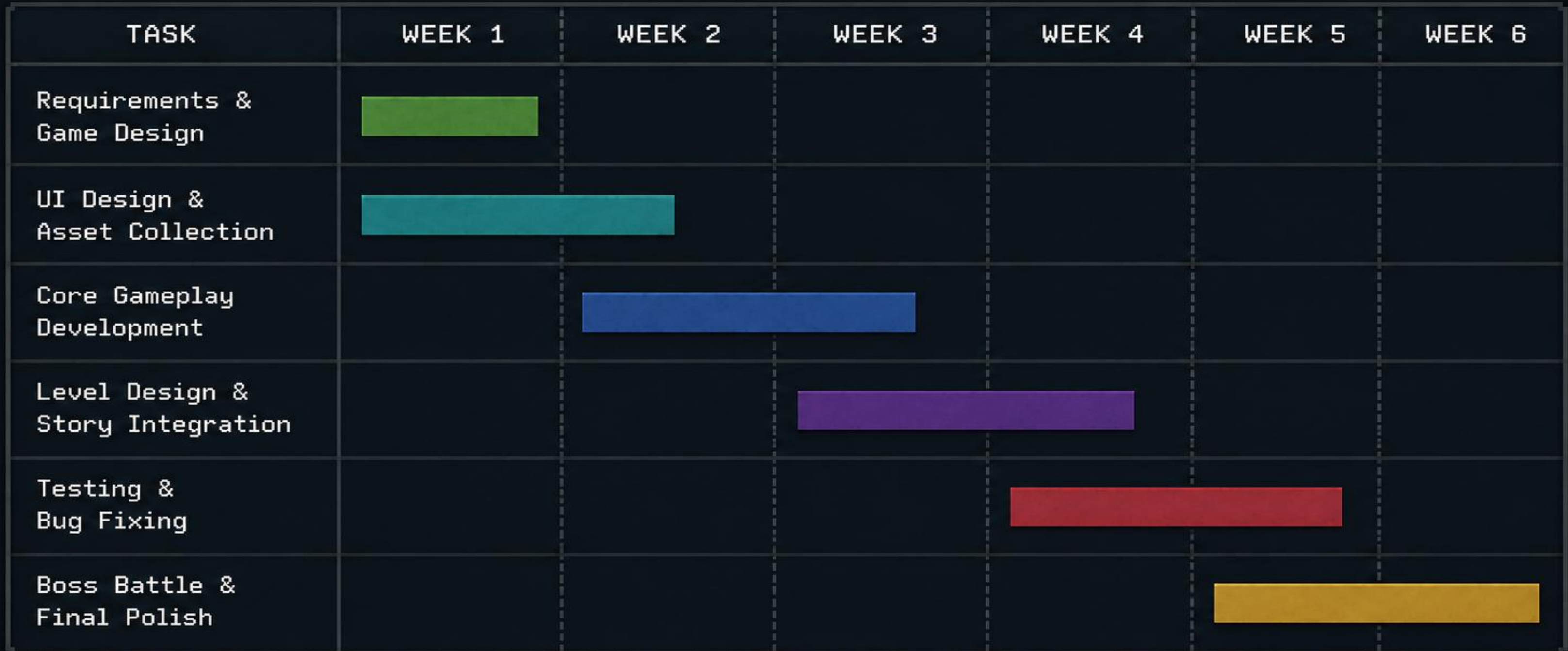
Version control and team collaboration



What Makes Us Different

Existing Game	What They Do	What We Do Differently
Plague Inc.	Strategy simulation — control a pathogen to infect the world	Real-time cooperative action on the ground, not strategy from above
Left 4 Dead	Co-op zombie shooter focused on combat and survival	Public health missions, not just combat — isolate, sanitize, rescue
Resident Evil	Horror survival with virus outbreaks as backdrop	Educates about outbreak response while being entertaining
Pandemic	Board game — turn-based cooperative strategy	Digital real-time experience with dynamic contamination
The Infected Hour	The only 2D co-op combining action with public health education	Emotional narrative depth through character backstory reveals

GANTT CHART



Game-Prototype



SDG 3: Good Health & Well-Being

3

GOOD HEALTH AND WELL-BEING

Target : Strengthen the capacity of all countries for early warning, risk reduction and management of national and global health risks

HOW OUR GAME CONTRIBUTES



Educates About Outbreak Response

Players learn real procedures used in health emergencies



Raises Sanitation Awareness

Highlights the importance of isolation and cleanliness



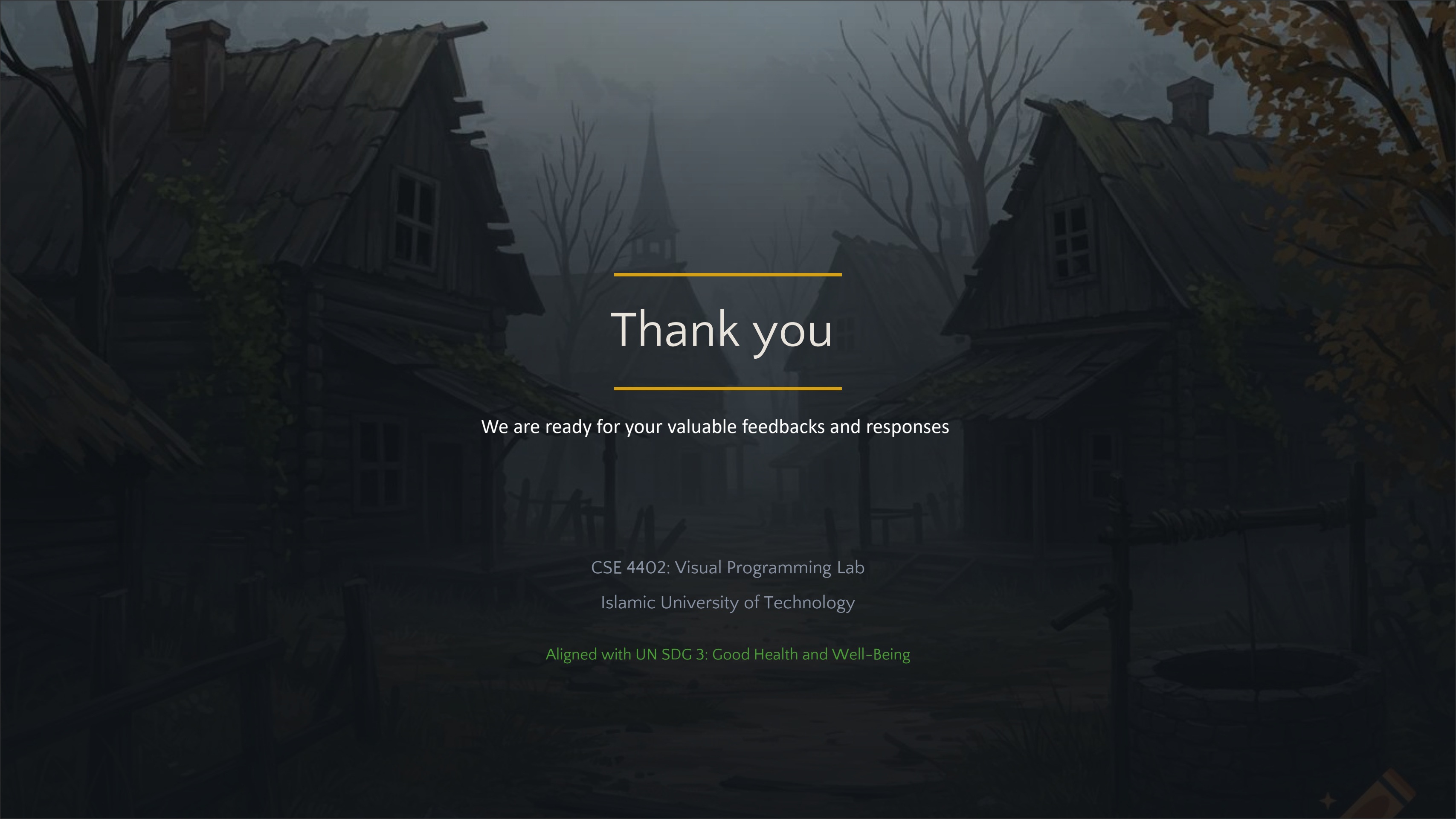
Demonstrates Cooperation Value

Shows how teamwork is critical in health emergencies



Makes Health Concepts Accessible

Interactive gameplay makes public health engaging



Thank you

We are ready for your valuable feedbacks and responses

CSE 4402: Visual Programming Lab

Islamic University of Technology

Aligned with UN SDG 3: Good Health and Well-Being